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RESEARCH REVIEW

The carrying capacity of ecosystems

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ABSTRACT

We analyse the concept of carrying capacity (CC), from populations to the biosphere, and offer a definition suitable for any level. For communities and ecosystems, the CC evokes density-dependence assumptions analogous to those of population dynamics. At the biosphere level, human CC is uncertain and dynamic, leading to apprehensive rather than practical conclusions. The term CC is widely used among ecological disciplines but remains vague and elusive. We propose the following definition: the CC is 'the limit of growth or development of each and all hierarchical levels of biological integration, beginning with the population, and shaped by processes and interdependent relationships between finite resources and the consumers of those resources'. The restrictions of the concept relate to the hierarchical approach. Emergent properties arise at each level, and environmental heterogeneity restrains the measurement and application of the CC. Because the CC entails a myriad of interrelated, ever-changing biotic and abiotic factors, it must not be assumed constant, if we are to derive more effective and realistic management schemes. At the ecosystem level, stability and resilience are dynamic components of the CC. Historical processes that help shape global biodiversity (e.g. continental drift, glaciations) are likely drivers of large-scale changes in the earth's CC. Finally, world population growth and consumption of resources by humanity will necessitate modifications to the paradigm of sustainable development, and demand a clear and fundamental understanding of how CC operates across all biological levels.

Keywords

Biosphere, community, ecosystem, hierarchy theory, population, renewable resource management, resilience, stability, sustainable development.

INTRODUCTION

Humans and human impacts on the biosphere continue to expand at an alarming rate. Unchecked, this will eventually lead to unsustainable degradation of the global biosphere. As early as 1798, Malthus raised the spectre of an exponential increase in human population but only arithmetic increase in available food (Fig. 1a). Verhulst (1838) modified the Malthusian model to include the concept of 'saturation level', termed 'carrying capacity' (CC), being the maximum population level that a given environment can support given finite resources (food, space, water, jobs, etc.), often termed 'logistic growth' (Fig. 1b). The biological rationale is simple. Theoretically, a new sexually reproducing population begins from two or more individuals. Initial near-exponential growth begins to slow as each new individual or unit of biomass causes a diminution in the environmental resources (food, space, etc.) available. Eventually, this demand provokes a decline in the actual numbers (or biomass) added to the population. As intraspecific competition intensifies, overall survival decreases, increasing selection pressure and stabilizing the total number of individuals or the total biomass at some constant maximum level (CC).

This idea is the cornerstone of the management of renewable resources (Hilborn *et al.*, 1995). For populations subject to commercial exploitation, the objective is usually to drive it to half CC, where the rate of increase is maximal (under a linear model). This level is a universal objective of different disciplines: Maximum Sustainable Yield in fisheries; Maximum Sustainable Cut or Annual Allowable Cuts in forestry (Robinson, 1993; Hilborn *et al.*, 1995); and 'economic CC' (most profitable) in livestock management (Dijkman, 1999). Conversely, if the goal is conservation of a population, then the 'ideal' end point is the CC. Finally, we can calculate the risk of extinction of a population (Punt, 2000) or determine how effectively a pest can be controlled, by how close its abundance lies to the origin of the curve. Given the actual conceptual gap between exploitation and

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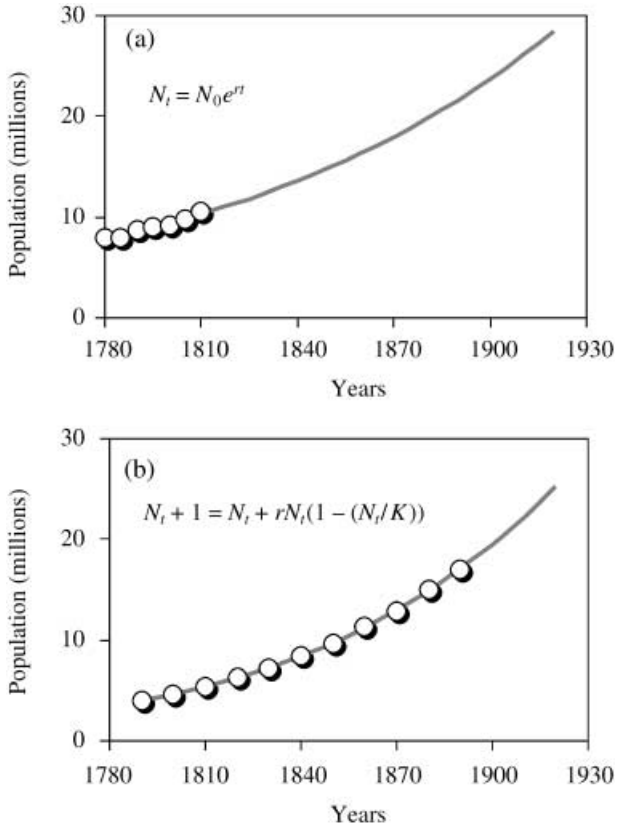

Figure 1 Fitting and forecast of the (a) Malthus (1826) and (b) Verhulst (1838) models of the population size data of England (1780–1810) and Belgium (1790–1890), respectively, with corresponding equations. Time step in (a) is yearly and in (b) is every five years.

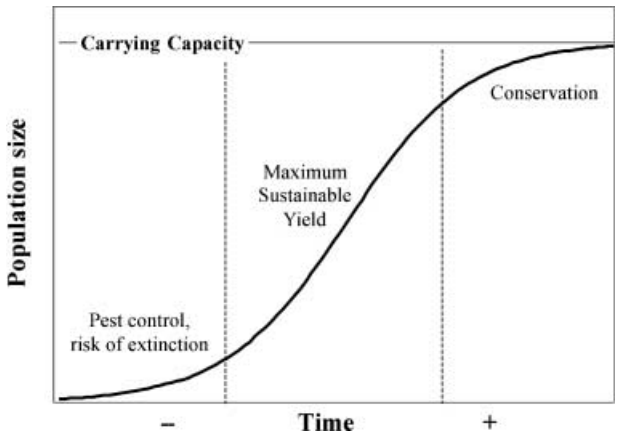

Figure 2 Schematic representation of three conceptual objectives (divided by vertical dotted lines) of renewable resource management, according to their relative position in the logistic population growth curve.

conservation (Fig. 2), their objectives necessarily counterbalance (Robinson, 1993; Struhasker, 1998).

The term CC is applied extensively, but imprecisely, among different scientific disciplines, from humans, to wildlife populations, to ecosystems and even to the whole Earth. Despite its fundamental significance and wide use, the concept remains controversial and vague (Catton, 1987). Here we adopt a hierarchical approach to propose and justify a more robust concept of CC, place it in the human and ecological context, and suggest how and to what point its application may be extended.

The carrying capacity of populations

Besides Verlhulst's logistic model, the CC can be expressed in any model that assumes density dependence. The stock-recruitment relationship (Beverton & Holt, 1957) for instance, supposes the reproductive output of a population increases logarithmically until births and deaths attain equilibrium. The CC is the point where the curve becomes asymptotic, also termed 'the equilibrium spawner abundance' (Myers *et al.*, 2001).

Since it is virtually impossible to consider all variables influencing the CC, it is commonly expressed as a function of certain fundamental resources. If the first trophic level is the ultimate source of food, then the CC can be estimated from the primary production. This measure is used in crop management (Tilman & Downing, 1994), livestock farming (Dijkman, 1999), aquaculture (Shuanglin *et al.*, 1998), fisheries (Pauly & Christensen, 1995; Vasconcellos & Gasalla, 2001) and even in marine mammal population dynamics (Schell, 2000). These models consider only the maximum population size. Nevertheless, the CC is a two-fold notion: the individuals (or biomass) and the factors that control their growth (Menczer, 1998). Hence, combining both elements would reflect more completely what the concept really represents.

In certain environments, space is the main determinant of the CC (Paine, 1966). Most environments may be subdivided as many times as there are existing classes of organisms (Hutchinson, 1959), making the measurement of the realized living space for a population a complicated task. Caddy & Stamatopoulos (1990) simplified the problem using fractal theory. They designed an artificial habitat with crevices that represent the disposition of spaces in a coral reef, and estimated the CC for some populations of infaunal species, expressed as the number of individuals of a given size per unit of irregularity in the habitat.

This duality has also been examined by integrating physical and biological models. Salvanes & Baliño (1998) formulated a coupled dynamic ecosystem model (DYNECOMAS) that integrates the topographical, meteorological, physical and biological characteristics that influence the production of a cod population (*Gaddus morhua*) in a fjord. Luo *et al.* (2001) combined a bioenergetics and water quality model to explore the spatio-temporal dynamics of the clupeid *Brevoortia tyrannus*. Another type of analysis that has enjoyed broad acceptance is ECOPATH (Christensen & Pauly, 1992), which incorporates tropho-dynamics elements of ecosystems to model the response of various populations in space and time (Walters *et al.*, 1997, 2000) to fishing pressure and environmental forcing. Christensen & Pauly (1992) explored these properties to determine the CC of a number of groups of marine organisms of commercial interest. The results of dual models are expressed in units of biomass, number or energy per unit of time, volume, area or food supply.

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Even species dynamics that are not governed by density dependence still have a CC. In this case, physical environmental factors exert the same potential effect on each member of the population, independently of how many there are (Sissenwine, 1984). This can result in accumulated biomass being sporadically reduced by changing environmental conditions, irrespective of the current density. Such intermittent behaviour is frequently observed in epidemics (Rhodes *et al.*, 1997; Grenfell *et al.*, 2002).

The carrying capacity of communities

A community can also seize the opportunities presented by a new environment. However, distinct from the population growth, which only involves the aggregate development of individuals, a community will undergo a set of sequential structural changes controlled by mutual interactions between the species and the environment.

The first steps of such 'succession' are distinguished by fast-growing and short-lived species whose production rates exceed the system respiration rate. As succession proceeds, these are replaced by others species with greater structural and functional complexity, that use the energy with progressive efficiency. This eventually results in a steady state, wherein respiration equals primary production (Tansley, 1935; Odum, 1969; Margalef, 1974; Christensen & Pauly, 1998). Per capita community productivity is very high during the initial phases of succession, and decreases progressively as an upper limit to biomass is reached (Margalef, 1974), and may be expressed as a sigmoid curve roughly akin to the logistic growth curve of a population (Hutchinson & Wollack, 1940), but marked by stages or 'steps' (Lindeman, 1942).

Is the community CC the maximum limit of succession? If particular species substitute progressively for others, then predecessor stages in the succession (seres), attain their CC and either remain restrained or eventually disappear. The species that succeed them will tend toward a different CC, each time closer to that of the ecosystem (Fig. 3). A similar idea (the 'maximum potential biomass'), developed independently, was proposed by Reichle *et al.* (1980) and supported by De Angelis (1980) and Taylor *et al.* (1990).

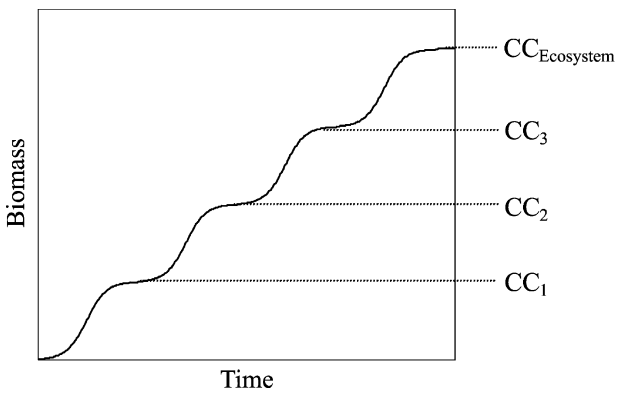

Figure 3 Theoretical development of the total biomass of an ecosystem during ecological succession. Each seral stage, represented as the steps in the curve, attains its own carrying capacity (CC, dotted lines) but with a general trend towards the ecosystem CC. Here are arbitrarily represented four seral stages.

Succession is recurrent and directional, and thus may be a predictable process. JABOWA (Botkin *et al.*, 1972) and SORTIE (Deutschman *et al.*, 1999) are examples of ecological succession models used in forestry. Other approaches based on ocean physics, thermodynamics and biological processes, such as ECOPATH, dynamic models (Robinson & Ware, 1994; Robinson, 1994) and spatially structured models (Henderson & Steele, 1996) like ERESM (Baretta *et al.*, 1995) have been applied to the spatio-temporal dynamics of aquatic communities. All of these models incorporate biomass data from primary producers, making it possible to obtain a representation of CC for any given community.

The carrying capacity of ecosystems

Communities appear in each step of succession via re-colonization by species from surrounding areas, which jointly determine overall ecosystem diversity. The natural history of this diversity establishes a link to CC, i.e. the maximum number of species the ecosystem can support. As for populations and communities, an ecosystem presents a finite resource base for its constituent species. If a habitat or adaptive zone (Futyyma, 1998) has relatively few species, immigration and diversification (including acquisition of evolutionary innovations (Benton, 1995), progress until the region attains a maximum supportable array of species over a given time. In fact, species diversification on evolutionary time scales has been likened to logistic population growth (Walker & Valentine, 1984). By analogy, the number of taxa equates to number of individuals or biomass, and the difference between rate of speciation and extinction, to the intrinsic growth rate. If, as the total number of taxa increases, the speciation rate decreases and extinction increases, equilibrium is eventually attained.

Biodiversity equilibria may be regulated by physical factors such as global climate and topographic changes (Futuyma, 1998), and the development of equilibria may be regulated by interactions between species. Once ecosystem biodiversity reaches saturation level, inter-specific interactions can lead to competitive exclusion of some taxa, and finer-scale niche partitioning (Lluch-Belda, 1978). The resources that govern ecosystem CC will place constraints on both the number of competitively superior species supported, and the number of divisions a single niche can withstand. Thus although biodiversity often appears to increase without approaching equilibrium, with new species potentially creating novel niches for others (Levinton, 1979; Maley, 1998), the same processes governing logistic-like growth of diversity must eventually constrain such exponential growth.

There is evidence suggestive of a global biodiversity CC, at least during certain periods in Earth's history (Miller & Sepkoski, 1988; Maley, 1998), e.g. from the Ordovician to the Permian (Courtillot & Gaudemer, 1996), but the general trend over the last 200 million years (My) has been an exponential increase in the total number of taxonomic families, with no obviously levelling off right up to the present (Dobzhansky, 1980; Futuyma, 1998; Maley, 1998; Benton, 2001).

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The carrying capacity of the biosphere

The earth is practically a closed system, and its resources are a finite base upon which human activities depend (Arrow *et al.*, 1995). Unceasing growth of world population, and the way ecosystems have been exploited to meet our consumption requirements, justifies a reconsideration of the concept of CC to plan our future course.

Following Malthus' (1798), concern over the expansion of the human population remained relatively latent for almost 150 years. Before 1960, world development was deeply slanted towards the economic domain, far from ecology, despite the fact that the bulk of raw materials fuelling economic activity were derived from the natural environment (Cruces, 1997). Reviving the Malthusian philosophy, Ehrlich (1968) and 'The Club of Rome' alerted public opinion to the possible consequences of world population growth and industrialization for food production and resource exhaustion. They concluded that within a century, the planet would reach the limits of growth, and advocated a mutually beneficial coupling of economics and ecology (Tamames, 1983). The Club of Rome, in turn, charged Prof D.L. Meadows' working group with carrying out an analysis of the threats to world society. The results, published in *The Limits to Growth* (Meadows *et al.*, 1972), were forceful and profound, spawning a global cult of 'environmentalism'. That same year, the United Nations convened the Stockholm Conference, with the aim of establishing an ecological foundation for management of the global environment. One of the results of this conference was the creation of the United Nations Environment Program (UNEP), which has promoted many of the global pro-environment initiatives (e.g. The Global Conservation Strategy and the Agenda 21), whose objectives are to reconcile long-term socio-economic and ecological stability through international cooperation and regulation.

Thirty years on, the most dire predictions of the new Malthusian philosophers have not been fulfilled. What Malthus never contemplated, and what his modern supporters underestimated, was humanity's capacity, through technological innovation, to increase our own CC (Trewavas, 2002), and that classical definitions of CC are not directly applicable to human ecology. The Earth's human CC depends upon complex interactions between a vast and often apparently unrelated gamut of conditions (Robinson, 1993; Cohen, 1995a), and cannot be characterized by a simple monotonic decline of the natural capital as the number of humans increase. However, this is not meant to imply the potential for perpetual human population growth. The present world population stands at around six billion people, and plausible estimates of the maximum number the planet can support range from 7.7 to 12 billion. Worryingly then, we are rapidly entering a zone where limits on the human CC of Earth have been anticipated (Cohen, 1995a; Fig. 4). Although there exists the technology and economic willingness to develop intensive, sustainable agriculture and other forms of food supply compatible with the aims of world development programs (Robinson, 1993; Tilman *et al.*, 2002), this cannot compensate indefinitely.

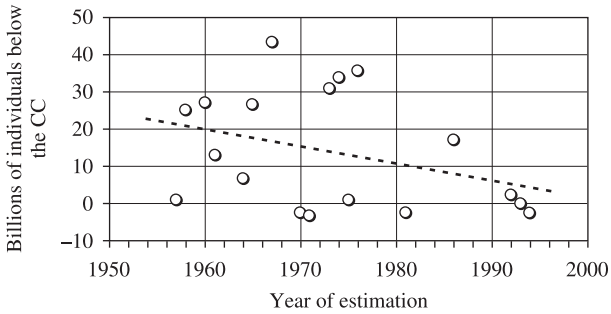

Figure 4 Difference between some estimates of Earth's human Carrying Capacity (CC) (Cohen, 1995b) and the world population data, from 1957 to 1994 (US Bureau of the Census, 2003), in billions of individuals. The negative linear trend (dotted line) suggests that the human population is approaching the Earth's CC as the differences between the two are getting smaller. Some authors argue that humans are already beyond sustainable population levels (negative values in 1970, 1971, 1981 and 1994). Time scale on the *X*-axis is discrete, according to the year when the estimates were made.

DISCUSSION

The CC: from definition to concept

Dhondt (1988) critiqued the different definitions and usages of the CC concept, and concluded the term should be 'avoided at all costs' because its interpretation was vague, contextual and the target of several contradictions. First, it has been applied rather arbitrarily to populations, associations (May, 1972; Caddy & Stamatopoulos, 1990), communities (Pimm, 1984; Tilman, 1996; McCann, 2000), ecosystems (De Angelis, 1980; Franks, 2001; Vejre *et al.*, 2001), and the biosphere (Cohen, 1995a). Yet many differences exist in respect of the way these levels of biological organization are measured and interpreted. Second, the classical definition of CC ignores interdependencies (*sensu* Cohen, 1995a) between organizational levels and the physical, biological and socio-economic processes that determine them (Menczer, 1998).

With these criticisms in mind, and in the spirit of advancing the concept, we therefore propose the following definition: the CC is '*the limit of growth or development of each and all hierarchical levels of biological integration, beginning with the population, and shaped by processes and interdependent relationships between finite resources and the consumers of those resources*'. In practise, these limits have commonly been measured instantaneously as individuals, biomass, number of species, etc., and dependent on those factors that determine such growth (e.g. area, volume, productivity, food, environmental changes, energy, etc.). Our definition is compatible with various competing concepts (e.g. Dhondt, 1988; Den Boer & Reddingius, 1996), and makes no assumption about what model may apply. A finite level of resources does not mean that they may not vary with time; resources do vary, but the limiting amount will continue to be fixed at each new moment of time. When abstracted in mathematical models, CC may be expressed as a fixed parameter (e.g.

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'*K*' in the logistic model), but this is not a requirement in all circumstances. Although Dhondt (1988) was justified in highlighting the historical vagueness and elusiveness of the definition and application of CC, the power of its conceptualization makes it one of the major contributors to theorization and progress in the life sciences (Mayr, 1997).

The nature of the concept

Determination of factors influencing CC is one of the hot topics of modern ecology (Oesterheid *et al.*, 1992). Predation, competition, parasites, disease, food supply, living space, environmental variability and the interactions among these are some of the ingredients.

Consider a species whose populations are found in both caves and in open areas. The caves dwellers experience virtually constant physical conditions, but their abundance can vary as drastically as those populations living under highly variable conditions in the open (Kane & Ryan, 1983). Apparently, even when populations live in physically 'stable' habitats, their interconnections with other more variable populations can induce synchronous changes in abundance between both habitats. Alternatively, Den Boer & Reddingius (1996) suggest such synchrony may have, at least partially, a hereditary basis. Ultimately, any response to environmental pressure has a genetic component; the problem lies in identifying, interpreting, compartmentalizing and quantifying it. The more complex such relationships become and the higher the level of integration, the greater the possibility of unexpected patterns emerging. The reverse argument is also true. If changes in the abundance of a marine population are provoked by fluctuations in its food supply, and these in turn respond to local water mass movements, the CC of this population will not depend only on the broader circulation system, even if these patterns influence local conditions.

Extensions of the concept

Emergent properties of the CC at higher levels of organization relate to the structure and functioning of ecosystems. Key concepts here are 'stability' (tendency of ecosystems to return to an equilibrium state after a perturbation; McCann, 2000) and 'resilience' (time required for stability to be achieved; Holling, 1973). Historically, these concepts have been the focus of intense debate (May, 1972; Macarthur, 1955; Elton, 1958; Pimm, 1984), particularly given their apparent positive correlation with diversity (King & Pimm, 1983; Tilman *et al.*, 1998; Tilman, 1999; McCann, 2000; Aoki & Mizushima, 2001). Since stability can be measured as total ecosystem biomass (Tilman & Downing, 1994; Tilman *et al.*, 1998), with perturbations defined as constraints to this biomass (Taylor *et al.*, 1990), it is rational to recognize these properties as spatial and temporal (resilience) components of CC, keeping in mind that ecosystem stability is not merely a response to environmental perturbations.

Ecosystems may be modified, both structurally and functionally, by natural as well as human-related activities. Some terrestrial ecosystems have suffered profound changes caused by the ENSO phenomenon, including the transformation of large infertile areas into arable land (Holmgren *et al.*, 2001). In the marine realm, Scheffer *et al.* (2001) detected recurrent changes (regime shifts) in the ecological attributes over the entire Pacific Ocean over the last 25 years, possibly due to global atmospheric and oceanic forcing. These examples provide well-documented cases of natural alternation between stable states, without always implying ecological impoverishment, and they indicate the frequencies at which the ecosystem-level CC may change (years and decades, respectively).

Beyond these scales of variation are found others that have no comparable modern parallels, yet have repeatedly transformed the natural history of the biosphere. The most significant changes to global biodiversity may result from processes spanning thousands or millions of years. For instance, the repeated advance and retreat of continental ice sheets (every 20–100 millenia), compressing and expanding climate zones, have driven the alternation of species' distributions over continents and oceans (Seltzer *et al.*, 2002). Mass extinctions are possibly the largest-scale periodic variations known for the CC (26–32 million years), associated with continental drift (Futuyma, 1998) and changes in global climate (Raup, 1986). These recurrent changes have promoted a continuous replacement of evolutionary lineages, with a tendency towards an ever-increasing biodiversity up to the present. One of the last major changes in the biosphere's CC, more than 50 My ago, may have been responsible for facilitating the diversification of mammals, opening an evolutionary channel that eventually led to the appearance (among many others species), of man.

The hierarchical approach in theory and practice

Defining boundaries of biological systems in space and time is inherently difficult. If these represent a continuum, then this implies a continuous CC. Yet since ancient times, the living world has been thought of as a set of discrete systems. For instance, Clark *et al.* (1975) found that both the growth of conifers in western North America and the abundance-distribution of albacore tuna (*Thunus alalunga*) are linked to large-scale atmospheric changes over the eastern North Pacific. If population dynamics of distinct ecosystems are affected by common environmental changes, then we may envision modification of the CC at the midpoint between the biosphere and the ecosystem. However, the ecosystem CC may equally be viewed from an evolutionary perspective, with structural changes due to habitat loss, population depletion, anthropogenic extinctions (Jackson *et al.*, 2001) or biodiversity increases driven by the establishment of exotic species (Sax & Gaines, 2003).

The relatively stable dominance structure of copepod associations off California over 49 years, despite significant alterations to current flows at different frequencies (Lluch *et al.*, 2001), implies a combination of hierarchies (Rebstock, 2001). The spatial changes involved were similar to those of the tunas and trees, and although this presumably affected the CC of several species in different ecosystems (Lluch *et al.*, 2003), it could not be detected, by means of structural composition, at lower levels of organization.

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Linking CC among levels

The relationship of CC among populations, communities and ecosystems changes depending on both the scale of measurement and emergent properties at each level. Approaches for determining CC at one level may not be useful at another, because of unexpected synergisms or antagonisms. Suppose total biomass is assumed to represent the CC. If some species exert a disproportional influence on the ecosystem relative to their numbers (Piraino & Fanelli, 1999), its CC will depend principally on their abundances. However, in the absence of dominant species, the CC estimated from species richness or composition would be more informative.

Determining the total animal, plant, fungal and microbial diversity in a tropical rain forest is perhaps as difficult as deriving biomass measures. A problem with using total diversity as a surrogate of CC is that when an ecosystem's original area of occupancy is drastically diminished, the biodiversity decrease may not be proportional (Mann, 1991), depending primarily on the distribution of geographically wide-ranging species vs. limited-range endemics. In fact, the variety, density and dynamism of tropical populations and communities make any robust measurement of ecosystem CC impracticable. Ecologists have simplified this problem by instead assessing the diversity of selected taxa, like plant communities. In some temperate ecosystems, species richness is often indicative of human-induced effects (Weber *et al.*, 2004), whilst in tropical ecosystems, relative abundance is used to reveal long-term ecosystem shifts (Bush *et al.*, 2004).

Tracing variations in CC with a single variable is practical but insufficient. For example, in the California Current Pelagic ecosystem, sardines and anchovy populations show negative covariance (Chavez *et al.*, 2003). Small pelagics are dominant, and their combined biomasses may be used as measure of ecosystem CC, but asynchronies mean that total biomass shows less variation than individual species' CC. Similar relationships are apparent from experiments on terrestrial ecosystems (Doak *et al.*, 1998; Tilman, 1999). Moreover, assuming that because of competition, sardine abundance mediates anchovy populations (Lluch *et al.*, 1991), the CC of the latter may be indirectly and inversely determined by physical processes affecting the former, such as turbulent mixing, wind fields and upwelling intensity. Primary productivity (biomass, pigment concentration) is commonly used to indicate the CC of marine ecosystems, because although its links with higher trophic levels are often nonlinear (Cury *et al.*, 2001), it nevertheless reflects the system's potential for biomass production.

Dominant species are more common in extratropical regions (e.g. pine forest, savanna) where environmental conditions show wider amplitude, and determine ecosystem CC via fluctuations in numbers and geographical ranges of key species. By contrast, tropical ecosystems (e.g. rain forest, coral reefs) are comparatively more constant, containing a multitude of species often with no clearly dominant taxa (Dobzhansky, 1950). Under relatively stable and benign physical conditions, competitive and symbiotic relations will determine, on ecological scales, the CC of populations and communities, and on evolutionary scales, the diversification of life. Natural selection will favour those species best able to subdivide common resources, enhancing ecosystem CC, rather than adaptive features promoted by intermittently favourable conditions (e.g. hibernation, large-scale migrations, wider trophic spectra). This may explain why tropical regions have more endemics (Dynesius & Jansson, 2000) and highly-specialized taxa.

The trophic pyramid provides another context for evaluating CC across hierarchical levels. Dominant species/groups may represent entire trophic levels, and depending on the mechanisms of energy control, modifications to their CC may be detectable at other levels as control from either: (a) bottom-up (primary producers to predators) (b) intermediate-level ('wasp-waist'), or (c) top-down (predators to primary producers through trophic cascades; Polis *et al.*, 2001). Although top predators may exert important influences on lower trophic levels, primary producers ultimately define the scope of ecosystems. Indeed, some of the great mass extinctions likely involved a reduction in primary production (Solé *et al.*, 2002). Some degree of bottom-up energy control always exists (Cury *et al.*, 2001), even when top-down effects predominate. In the absence of dominant taxa, however, variations in the CC of particular populations, irrespective of trophic level, may be sufficiently diluted through the food web to be imperceptible at the community or ecosystem level.

Carrying capacity and resource management

It took 120 years before the concept of CC was applied directly to a concrete problem (Pearl & Reed, 1920). The results were so illustrative that the idea spread rapidly to many diverse scientific disciplines (Tsoularis, 2001), but particularly in ecology. The importance of the concept, however, was not always appreciated — not due to difficulties in recognizing the basic definition, but because the concept was neither understood nor applied in its fullest sense. Despite the pluralistic and dynamic nature of the CC (Margalef, 1991), there still is a strong reluctance to recognize its ubiquity in natural systems, especially in vexed resource management issues (Pauly *et al.*, 2002, 2003).

The CC is commonly assumed to be constant in population dynamics models used for resource assessment. Although such simplification may not be critical to short-term studies, the fact that natural systems always exhibit some inherent variability (Hilborn & Ludwig, 1993) means such deterministic models lose robustness over the long term (Lluch-Belda *et al.*, 1999). The panorama is not markedly different at higher levels of organization. Definitions of stability and resilience, applied to communities and ecosystems (Tilman *et al.*, 2002), assume static underlying system properties. Even when different states of equilibrium are considered, each is assumed to be fixed (Gunderson, 2000). We believe it would be more profitable to consider domains of stability (limited ranges over which the CC varies) as naturally variable and dynamic, at all hierarchical levels.

Many exploited ecosystems are in advanced stages of succession (a relatively stable state with low resilience), dominated by species with low productivity/biomass ratios (P/B). If maximal yield is the primary management goal, an acceptable option

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may be to focus exploitation on naturally immature ecosystems (Odum, 1969) subject to intermittent nutrient supplies. These are comparatively more productive systems characterized by species with high P/B and higher resilience. Alternatively, exploitation of mature systems could be improved by rewarding sustainable practices, such as increasing efficiency of fertilizers and regulating the use of pesticides and antibiotics (Tilman *et al.*, 2002). Of course, high stability and diversity does not always mean low resilience. Some temperate terrestrial ecosystems show relatively variable biological and physical conditions, yet are exploited by a diversity of taxa whose integrated effect is to confer high ecosystem resilience (Holling, 1973). Yet if stability, resilience and diversity are correlated positively, then the potential for exploitation of any ecosystem would be restricted. This complexity means management strategies focusing only on diversity tend to be inadequate.

Alterations of the CC (e.g. via climatic fluctuations; Hare *et al.*, 1999) may affect even the best-protected species. Between 1999 and 2000, more than 630 grey whale (*Eschrichtius robustus*) carcasses were found along the North American coastline. Speculations about this beaching focused upon illness, starvation, human activity or some combination of these, suggesting a population close to CC (Moore *et al.*, 2001), i.e. eliminating the surplus. The northern American grizzly bear (*Ursus arctos*) is another interesting case. For over 25 years, the main working hypothesis was that hunting and habitat loss were the main causes of mortality. An alternative hypothesis is that its low reproductive rate was a response to a coupling between the population dynamics and changes in food production, provoked, in turn, by climate shifts (Picton *et al.*, 1986). There are other similar examples from reptiles, birds and mammals addressing natural fluctuations (changing CC) of non-exploited populations (Wikelski & Thom, 2000; Forchhammer *et al.*, 2001; Croxall *et al.*, 2002).

If no other reference point besides an equilibrium is considered, situations like these may be interpreted as 'resource crises' (Gunderson, 2000), when the root cause is environmental change. Whether part of the natural variability, or derived from human-induced effects on global and regional climate (Hulme *et al.*, 1999; Shindell *et al.*, 1999), these situations nonetheless represent changes in the CC. Given our present inability to predict accurately these events, the adoption of adaptive management offers an attractive way forward (Hilborn & Ludwig, 1993; Dijkman, 1999). We must assume that surprises are inevitable in variable natural systems; human interactions with ecosystems will only increase, and thus our knowledge will always be incomplete. When confronting new or changing situations, we must be prepared to 'tune' our responses rather than committing to rigid, unwavering traditional management strategies.

Carrying capacity and sustainable development

The fundamentals of classical sustainable development entail maximum use of a resource without damaging the system's regenerative capacity (Lluch-Belda *et al.*, 1999). Yet because understanding of how living systems behave remains rudimentary, a precautionary philosophy prevails: when uncertainty exists, prudence is required (Goodland, 1995), even if maximum sustainable yields must be forfeited. The 2002 World Summit on Sustainable Development (WSSD) nominated its top priority to be the eradication of hunger and poverty worldwide. It is therefore inevitable that the demand for natural resources will continue to grow, and may ultimately force a fundamental revision of the spirit of sustainable development. Should the final balance tend towards maximum use, it may be possible to satisfy the expectations that feeding everybody assumes. The logistical capacity of humankind exists to achieve this, although the natural bases may not support such sustained exploitation indefinitely.

There is some cause for optimism. Some American and many European countries have successfully lowered or halted their rates of growth (Haub, 1995), and strong efforts are being made to provide women worldwide with the necessary knowledge and resources to make informed reproductive choices (Harte, 1996). There is also much underused capacity for food production and a tremendous amount of biodiversity that has not been used to satisfy immediate human needs (MacDonald, 1992). Conversely, many regions are already heavily overexploited and degraded, and problems of distribution and access continue to stymie the equitable distribution of food and essential services (Kates, 1998). Thus, whilst the means certainly exist to support a higher human CC (Robinson, 1993; Tilman *et al.*, 2002; Trewavas, 2002), the path towards truly sustainable global development will be a difficult one to tread, and will rely on the intelligent use of a finite and variable set of resources.

CONCLUSIONS

1 The notion that resource limitation must eventually constrain the growth of populations is appealing. The most common frame of analysis has been to assume density-dependence, wherein it acquires a name (CC) and a practical meaning. Estimation of CC will always be limited, because the environmental matrix is heterogeneous, but qualitative and quantitative accuracy can be maximized by establishing causal relationships between populations and their environment.

2 At the community level, species substitute each other, seizing with an increasing efficiency the opportunities that the environment provides until respiration equals primary production. Such ecological succession implies a CC for each kind of community, eventually reaching an ecosystem CC determined by the biotic and physical environment.

3 Over evolutionary time scales, there also seems to be a limit to the total number of species an ecosystem can maintain. The processes are analogous. The relationship between speciation and extinction rates, regulated by environmental conditions and biological interactions, corresponds to density dependent effects, until equilibrium (CC) is attained. Although such limits seem to have occurred during certain periods in the past, the trend over the past 200 My has been an exponentially increasing global diversity, as large-scale changes in the physical environment continually opened new evolutionary opportunities.

4 In the human context, rates of consumption and production may ultimately balance, as population size increases and natural

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capital is depleted, invoking the anthropogenic CC first envisaged by Malthus (1798).

5 The original definition of CC has lost meaning as it has become more generally used. We propose the following concept: 'the limit of growth or development of each and all hierarchical levels of biological integration, beginning with the population, and shaped by processes and interdependent relationships between finite resources and the consumers of those resources'. This concept addresses the bonds between biotic and physical components whilst acknowledging temporal and spatial variability (non-equilibrium).

6 If total biomass is used to evaluate ecosystem stability and resilience, then these properties might be considered dynamic components of CC. Commercial exploitation may focus more profitably on naturally immature ecosystems, and the utilization of comparatively more mature systems can be improved by rewarding sustainable practices.

7 If the CC of key species change, then the CC of higher hierarchical levels may also change, whether in terms of biological integration or trophic structure. In the former, such alteration affects the community/ecosystem CC, whilst in the latter, the repercussions are felt across the trophic hierarchy. Without dominant species, modification of a single population's CC will only influence closely-related populations and not the CC or structure of higher hierarchical levels.

8 Natural changes in CC necessarily reflect changes in biological–physical attributes. Therefore, management schemes that assume constant CC, or ignore its relationship with natural or human-induced environmental changes, lose robustness and generality.

9 Adaptive management is the most widely accepted solution for confronting the unpredictability of renewable resources. Natural resource management must consider the ever-changing interaction between physical and biological systems, and react according to acquired experience and historic knowledge in a continuous, iterative learning process.

10 Finally, the current ethos of 'sustainable development' is slanted towards preservation of the replacement capability of natural systems, rather than maximum use. However, the unceasing growth of world population may eventually bring equilibrium between the two. To face this future with confidence, humanity must endeavour to maintain the Earth's CC at a productive yet sustainable level, through improved logistical foundations, a more cooperative political climate, and better scientific understanding.

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